Supplementary Material: Zero-Sample Fault Diagnosis for EHSS: A Hierarchical Multi-Source Hybrid PI-RNN Based Method

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S1. Complete Model Supplement and Parameters

This supplementary material provides the detailed physical model of the electrohydraulic servo system (EHSS) presented in the main manuscript. It specifies the function for friction, leakage, valve flow, and electromagnetic actuation, together with the governing state equations and numerical implementation. Key physical parameters are summarized in Table S1.

a) Friction, flow force, and leakage.: For $j \in \{s, c\}$, define $t_j = \tanh(\dot{x}_j / v_{\epsilon,j})$ and $h_j = F_{Cj} + F_{Sj} \exp[-(\dot{x}_j / v_{Sj})^2]$. The signed friction forces are

$$\begin{aligned} F_{\mathrm{fric},s} &= -B_s \dot{x}_s - h_s t_s, \\ F_{\mathrm{fric},c} &= B_c \dot{x}_c + h_c t_c + B_{c,\mathrm{asym}} \dot{x}_c t_c \\ &\quad + F_{Cc,\mathrm{asym}} t_c^2. \end{aligned} \tag{S1}$$

The spool flow force and cross-chamber leakage are

$$\begin{aligned} F_{\mathrm{flow},s} = -\Bigg[& c_{F,1} \frac{P_A - P_B}{P_{\mathrm{ref}}} + c_{F,2} \frac{P_A + P_B}{P_{\mathrm{ref}}} \\ & + c_{F,3} \frac{P_A^2 - P_B^2}{P_{\mathrm{ref}}^2} \Bigg], \end{aligned} \tag{S2}$$

$$Q_{L,A} = -Q_{L,B} = C_\ell (P_A - P_B). \tag{S3}$$

b) Metering areas and discharge coefficients.: Let $[z]_+ = \max(z, 0)$ and $\mathcal{A}(w, z) = \min\{A_{\max}, w \min(\ell_u, [z]_+)\}$. The four metering areas are

$$\begin{aligned} A_A &= \mathcal{A}(w_{SA}, x_s / s_x - d_A), \\ A_{A2} &= \mathcal{A}(w_{TA}, -x_s / s_x - d_A), \\ A_B &= \mathcal{A}(w_{SB}, -x_s / s_x - d_B), \\ A_{B2} &= \mathcal{A}(w_{TB}, x_s / s_x - d_B). \end{aligned} \tag{S4}$$

For $j \in \{A, B\}$, both edges use

$$\begin{aligned} C_{d,j} &= \mathrm{clip}_{[C_{d,\min}, C_{d0}]} \Big\{ C_{d0} \big(1 + k_o \chi \\ &\quad + k_{Re} \frac{Re_j}{Re_0 + Re_j} \big) \Big\}, \\ \chi &= \min\left(\frac{|x_s|}{s_x \ell_u}, 1 \right), \quad Re_j = \frac{4\rho |Q_j|}{\pi \mu D_h}. \end{aligned} \tag{S5}$$

With $\delta = \Delta P / P_{\mathrm{ref}}$ and $\varepsilon = \epsilon_p / P_{\mathrm{ref}}$, the signed pressure-flow map is

$$\mathcal{R}_{n_q}(\Delta P) = \sqrt{\frac{2P_{\mathrm{ref}}}{\rho}}\, \delta (\delta^2 + \varepsilon^2)^{n_q/4 - 1/2}. \tag{S6}$$

c) Mechanical stops.: With $v_s = \dot{x}_s$ and $v_c = \dot{x}_c$, the signed stop forces are

$$\begin{aligned} S_s = & -k_{\mathrm{stop},s} \operatorname{sign}(x_s) [|x_s| - x_{s,\max}]_+ \\ & - b_{\mathrm{stop},s} v_s \mathbf{1}_{|x_s| > x_{s,\max}}, \\ S_c = & -k_{\mathrm{stop},c} [x_c - \mathrm{clip}_{[0,\ell_c]}(x_c)] \\ & - b_{\mathrm{stop},c} \big[\min(v_c, 0) \mathbf{1}_{x_c < 0} \\ & + \max(v_c, 0) \mathbf{1}_{x_c > \ell_c} \big]. \end{aligned} \tag{S7}$$

The unloaded simulation uses $F_{\mathrm{ext}} = 0$.

d) Electrical and electromagnetic fits.: Let $i_e = \mathrm{clip}_{[i_{\min}, i_{\max}]}(i)$ and $g_e = \mathrm{clip}_{[0, g_{\max}]}(g_0 - x_s)$. The FEM-fitted electromagnetic relations are

$$\begin{aligned} \Phi_0 &= a(i_e) + g_e b(i_e) + g_e^2 c(i_e), \\ \lambda &= N_c \Phi_0, \qquad F_{em} = \sum_{p,q} e_{pq} i_e^p \Phi_0^q, \\ a(z) &= \sum_{n=0}^{9} a_{\Phi,n} z^n, \quad b(z) = \sum_{n=0}^{8} b_{\Phi,n} z^n, \\ c(z) &= \sum_{n=0}^{7} c_{\Phi,n} z^n, \\ \lambda_i &= N_c [a'(i_e) + g_e b'(i_e) + g_e^2 c'(i_e)], \\ \lambda_x &= -N_c [b(i_e) + 2 g_e c(i_e)] \\ &\quad \times \mathbf{1}_{0 < g_0 - x_s < g_{\max}}. \end{aligned} \tag{S8}$$

The coefficients $a_{\Phi,n}$, $b_{\Phi,n}$, $c_{\Phi,n}$, and e_{pq} are given in the accompanying parameter file; unspecified coefficients are zero. All remaining numerical settings are listed in Table S1.

S2. Convolution-kernel design and disturbance verification

The EHSS is characterized by a cascaded physical structure, following $u \to i \to x_s \to \{P_A, P_B\} \to x_c \to F_{\mathrm{ext}}$. Based on these physical dependencies, the convolutional kernels $\boldsymbol{K}_S$ and $\boldsymbol{K}_C$ are designed through a physics-constrained optimization procedure, as detailed below.

The main structure is shown in the Figure S1. The three physical

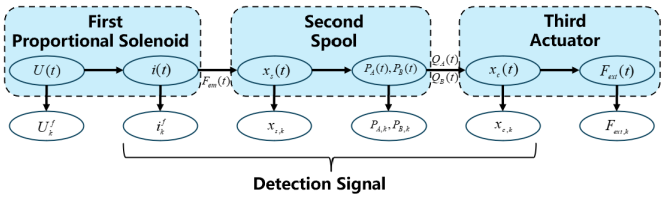


Fig. S1: Hierarchical Structure and Signal Definitions.

stages are the proportional-solenoid stage, the spool stage, and the hydraulic-actuator stage. They describe subsystem decomposition and are distinct from the signal nodes used in the hierarchical residual construction. The four internal monitored hierarchy levels are

$$\mathcal{H}_1 = \{i\}, \qquad \mathcal{H}_2 = \{x_s\}, \qquad \mathcal{H}_3 = \{\boldsymbol{P}\}, \qquad \mathcal{H}_4 = \{x_c\}, \quad \boldsymbol{P} = [P_A, P_B]^{\mathrm{T}}. \tag{S9}$$

The input u and the load variable F_{ext} are non-diagnostic boundary/context nodes. Hence, the grouped topological sequence used for hierarchical inference is

$$[u,\ i,\ x_s,\ \boldsymbol{P},\ x_c,\ F_{\mathrm{ext}}], \qquad N_h = 6, \quad \mathcal{V} = \{1, \ldots, N_h\}, \quad \mathcal{V}_M = \{2, \ldots, 5\}. \tag{S10}$$

Here, $N_h = 6$ counts grouped topological nodes, whereas N_c is reserved for the number of solenoid coil turns. The set $\mathcal{V}$ denotes hierarchy nodes and is distinct from the diagnostic indicator $\mathcal{I}$. The four internal nodes contain five monitored scalar channels,

Table S1: Constitutive-model and simulation settings (SI units). Values in parentheses follow the parameter order.

Parameters	Values	Parameters	Values
(B_s, B_c)	$(2.46, 8.96\times10^{3})$ N s/m	(w_{SA}, w_{TA})	$(13.9, 11.6)\times10^{-3}$ m
$B_{c,\mathrm{asym}}$	6.68×10^{3} N s/m	(w_{SB}, w_{TB})	$(6.92, 11.3)\times10^{-3}$ m
(F_{Cs}, F_{Ss})	$(1.17, 6.09)\times10^{-3}$ N	(d_A, d_B)	$(7.38, 6.45)\times10^{-6}$ m
(F_{Cc}, F_{Sc})	$(746, 97.5)$ N	$(A_{\max}, \ell_u)$	$(1.13\times10^{-5}\,\mathrm{m}^2, 4.92\times10^{-4}\,\mathrm{m})$
$F_{Cc,\mathrm{asym}}$	702 N	s_x	1
$(v_{Ss}, v_{\epsilon,s})$	$(0.0105, 0.00243)$ m/s	$(C_{d,\min}, C_{d0})$	$(0.300, 0.650)$
$(v_{Sc}, v_{\epsilon,c})$	$(0.00729, 0.00851)$ m/s	(k_o, k_{Re})	$(-0.397, 0.169)$
$(c_{F,1}, c_{F,2}, c_{F,3})$	$(0.0378, 0.0568, -0.0253)$ N	(Re_0, D_h)	$(2.00\times10^{3}, 0.0120\,\mathrm{m})$
P_{ref}	6.89×10^{6} Pa	(n_q, ϵ_P)	$(0.703, 1.00\times10^{4}\,\mathrm{Pa})$
C_ℓ	8.49×10^{-13} m^3/(s Pa)	$x_{s,\max}$	2.00×10^{-3} m
$k_{\mathrm{stop},s}$	2.00×10^{6} N/m	$k_{\mathrm{stop},c}$	1.50×10^{8} N/m
$b_{\mathrm{stop},s}$	1.00×10^{2} N s/m	$b_{\mathrm{stop},c}$	9.95×10^{4} N s/m
$(i_{\min}, i_{\max})$	$(0.800, 2.85)$ A	$(g_0, g_{\max})$	$(2.12, 4.00)\times10^{-4}$ m
N_c	600	F_{ext}	0 N

$[i, x_s, P_A, P_B, x_c]^{\mathrm{T}}$, because the pressure node contains two chamber-pressure components. In particular, P_A and P_B are parallel components of the same hydraulic level; neither is upstream or downstream of the other.

A. Kernel-design procedure

At each hierarchy level, three types of information are available: upstream inference, PI-RNN prediction, and downstream inference. The pressure pair (P_A, P_B) occupies one physical level. As in the manuscript, the normalized deviation matrix is formed separately for pressure branches $b \in \{A, B\}$; the same 3×3 kernels are applied to both branches. After saturation and threshold processing, it is written as

$$\bar{\boldsymbol{D}}^{(b)}_{\nu,k} = \frac{a_{\mathrm{sat}}|\boldsymbol{D}^{(b)}_{\nu,k}|}{a_{\mathrm{sat}} + |\boldsymbol{D}^{(b)}_{\nu,k}|}, \tag{S11}$$

$$\widetilde{\boldsymbol{D}}^{(b)}_{\nu,k}(i,j) = \begin{cases} \bar{\boldsymbol{D}}^{(b)}_{\nu,k}(i,j), & \bar{\boldsymbol{D}}^{(b)}_{\nu,k}(i,j) > \frac{a_{\mathrm{sat}}}{a_{\mathrm{sat}}+1}, \\ 0, & \bar{\boldsymbol{D}}^{(b)}_{\nu,k}(i,j) \le \frac{a_{\mathrm{sat}}}{a_{\mathrm{sat}}+1}. \end{cases} \tag{S12}$$

Here $a_{\mathrm{sat}} = 4$, so the activation cutoff is 0.8. For each branch, valid convolution with unit stride and no padding extracts four local residual windows:

$$\mathbf{U}^{(b)}_{\nu,k,l} = \widetilde{\boldsymbol{D}}^{(b)}_{\nu,k}(l : l+2, :) \in \mathbb{R}^{3\times3}, \qquad l = 1, \ldots, 4. \tag{S13}$$

The branch-specific diagnostic responses are defined by

$$z^{(b)}_{S,k,l} = \langle \mathbf{U}^{(b)}_{S,k,l}, \boldsymbol{K}_S \rangle, \qquad z^{(b)}_{C,k,l} = \langle \mathbf{U}^{(b)}_{C,k,l}, \boldsymbol{K}_C \rangle, \tag{S14}$$

where $\boldsymbol{K}_S$ is the sensor-fault kernel, $\boldsymbol{K}_C$ is the component-propagation kernel, and $\langle\cdot,\cdot\rangle$ denotes the Frobenius inner product. For $\nu \in \{S, C\}$, define $\bar{z}_{\nu,k,l} = (z^{(A)}_{\nu,k,l} + z^{(B)}_{\nu,k,l})/2$. The two pressure branches are fused exactly as in the manuscript:

$$\mathbf{z}_{\nu,k} = [\bar{z}_{\nu,k,1}, \bar{z}_{\nu,k,2}, z^{(A)}_{\nu,k,3}, z^{(B)}_{\nu,k,3}, \bar{z}_{\nu,k,4}]^{\mathrm{T}}. \tag{S15}$$

Its five entries correspond to $[i, x_s, P_A, P_B, x_c]^{\mathrm{T}}$. The first, second, and fifth responses are averaged across branches, while the two pressure responses remain separate. In the design inequalities below, $z^S_{k,l}$ and $z^C_{k,l}$ denote the entries of these final fused indicators, with $l = 1, \ldots, 5$.

The kernel optimization starts from the physical residual patterns of the EHSS. For a sensor fault, the physical state is unchanged, while the local measurement becomes inconsistent with upstream inference, PI-RNN prediction, and downstream inference. Thus, the expected sensor-fault residual topology is

$$\mathbf{M}_S = \begin{bmatrix} 0 & 0 & + \\ + & + & + \\ + & 0 & 0 \end{bmatrix}. \tag{S16}$$

For a component fault, the physical state changes and the residual propagates downstream along the causal chain. Thus, the expected component-fault residual topology is

$$\mathbf{M}_C = \begin{bmatrix} 0 & 0 & + \\ + & + & 0 \\ 0 & + & 0 \end{bmatrix}. \tag{S17}$$

Based on these two physical templates, paired A/B-branch residual-pattern datasets are generated for kernel selection. In the following dataset notation, $\mathbf{W}$ represents the branch-window pair $\{\mathbf{W}^{(A)}, \mathbf{W}^{(B)}\}$ and its associated fused five-position response:

$$\mathcal{D}_N = \{\mathbf{W}^N_{k,l}\}, \tag{S18}$$

$$\mathcal{D}_S = \{(\mathbf{W}^S_{k,l}, l^*)\}, \tag{S19}$$

$$\mathcal{D}_C = \{(\mathbf{W}^C_{k,l}, l^*)\}, \tag{S20}$$

where $\mathcal{D}_N$, $\mathcal{D}_S$, and $\mathcal{D}_C$ denote normal, sensor-fault, and component-fault residual windows, respectively, and l^* is the true fault layer.

The optimization objective is to find two kernels that satisfy the following diagnostic inequalities.

For normal samples:

$$z^S_{k,l} < \tau_S, \qquad z^C_{k,l} < \tau_C, \qquad \forall l. \tag{S21}$$

For a sensor fault at layer l^*:

$$z^S_{k,l^*} > \tau_S, \tag{S22}$$

$$z^S_{k,l} < \tau_S, \qquad l \neq l^*, \tag{S23}$$

$$z^C_{k,l} < \tau_C, \qquad \forall l. \tag{S24}$$

For a component fault starting at layer l^*:

$$z^C_{k,l} > \tau_C, \qquad l \ge l^*, \tag{S25}$$

$$z^C_{k,l} < \tau_C, \qquad l < l^*, \tag{S26}$$

$$z^S_{k,l} < \tau_S, \qquad \forall l. \tag{S27}$$

In this work, the thresholds are set as

$$\tau_S = 4, \qquad \tau_C = 5. \tag{S28}$$

The above inequalities are converted into a hinge-loss optimization problem:

$$\begin{aligned} \min_{\boldsymbol{K}_S, \boldsymbol{K}_C} \quad & \mathcal{L}_N + \mathcal{L}_S + \mathcal{L}_C + \lambda_1 \left(\|\boldsymbol{K}_S\|_1 + \|\boldsymbol{K}_C\|_1\right) \\ & + \lambda_2 \left(\|\boldsymbol{K}_S\|_F^2 + \|\boldsymbol{K}_C\|_F^2\right), \end{aligned} \tag{S29}$$

where the normal-condition loss is

$$\mathcal{L}_N = \sum_{\mathbf{W}\in\mathcal{D}_N} \sum_l \left[z^S_{k,l} - \tau_S + \eta_S\right]_+^2 + \left[z^C_{k,l} - \tau_C + \eta_C\right]_+^2. \tag{S30}$$

The sensor-fault loss is

$$\begin{aligned}\mathcal{L}_S = &\sum_{(\mathbf{W},l^*)\in\mathcal{D}_S}\left[\tau_S+\eta_S-z^S_{k,l^*}\right]_+^2\\ &+\sum_{(\mathbf{W},l^*)\in\mathcal{D}_S}\sum_{l\neq l^*}\left[z^S_{k,l}-\tau_S+\eta_S\right]_+^2\\ &+\sum_{(\mathbf{W},l^*)\in\mathcal{D}_S}\sum_{l}\left[z^C_{k,l}-\tau_C+\eta_C\right]_+^2.\end{aligned}\tag{S31}$$

The component-fault loss is

$$\begin{aligned}\mathcal{L}_C = &\sum_{(\mathbf{W},l^*)\in\mathcal{D}_C}\sum_{l\geq l^*}\left[\tau_C+\eta_C-z^C_{k,l}\right]_+^2\\ &+\sum_{(\mathbf{W},l^*)\in\mathcal{D}_C}\sum_{l<l^*}\left[z^C_{k,l}-\tau_C+\eta_C\right]_+^2\\ &+\sum_{(\mathbf{W},l^*)\in\mathcal{D}_C}\sum_{l}\left[z^S_{k,l}-\tau_S+\eta_S\right]_+^2.\end{aligned}\tag{S32}$$

Here $[x]_+ = \max(x,0)$, and η_S, η_C are safety margins that force the responses to stay away from the decision thresholds.

To avoid overfitting and maintain physical interpretability, the kernel coefficients are constrained to be bounded half-integers:

$$K^S_{ij}, K^C_{ij} \in \left\{\frac{m}{2} \mid m\in\mathbb{Z}\right\},\tag{S33}$$

$$-\kappa \leq K^S_{ij}, K^C_{ij} \leq \kappa.\tag{S34}$$

This constraint prevents the diagnostic head from becoming a black-box classifier and ensures that each coefficient has a clear physical meaning.

To account for occasional disturbances, a bounded perturbation term is also considered:

$$\|\Delta\mathbf{W}\|_\infty \leq \epsilon.\tag{S35}$$

For non-target responses, robustness requires

$$\max_{\|\Delta\mathbf{W}\|_\infty\leq\epsilon}\langle\mathbf{W}+\Delta\mathbf{W},\mathbf{K}\rangle \leq \gamma-\eta.\tag{S36}$$

Equivalently,

$$\langle\mathbf{W},\mathbf{K}\rangle + \epsilon\|\mathbf{K}\|_1 \leq \gamma-\eta.\tag{S37}$$

For target weak faults, robustness requires

$$\min_{\|\Delta\mathbf{W}\|_\infty\leq\epsilon}\langle\mathbf{W}+\Delta\mathbf{W},\mathbf{K}\rangle \geq \gamma+\eta,\tag{S38}$$

which is equivalent to

$$\langle\mathbf{W},\mathbf{K}\rangle - \epsilon\|\mathbf{K}\|_1 \geq \gamma+\eta.\tag{S39}$$

The practical optimization procedure is summarized as follows:

1) Construct normal, sensor-fault, and component-fault residual-pattern datasets according to the physical propagation rules of the EHSS.
2) Apply saturation and threshold normalization to both $\widetilde{\boldsymbol{D}}^{(A)}_{\nu,k}$ and $\widetilde{\boldsymbol{D}}^{(B)}_{\nu,k}$.
3) Extract four local windows $\mathbf{W}^{(b)}_{k,l}$ in each pressure branch.
4) Convolve both branches with the shared kernels, then fuse the responses into the five monitored positions $[i, x_s, P_A, P_B, x_c]^{\mathrm{T}}$.
5) Optimize $\boldsymbol{K}_S$ and $\boldsymbol{K}_C$ under classification, margin, robustness, bounded-weight, and half-integer constraints.
6) Select the kernel pair that satisfies all diagnostic inequalities while preserving physical interpretability.
7) Validate the selected kernels on normal, sensor-fault, and component-fault residual datasets.

Following the above optimization process, the final diagnostic kernels are selected as

$$\boldsymbol{K}_S = \begin{bmatrix}-0.5 & -0.5 & 1\\ 1 & 1 & 1\\ 2 & -4 & 0\end{bmatrix},\qquad \boldsymbol{K}_C = \begin{bmatrix}-1 & 0 & 1\\ 1 & 3 & -3\\ -2 & 2 & 0\end{bmatrix}.\tag{S40}$$

The boundary entries for the sensor and component branches are denoted by $\zeta_S = 1$ and $\zeta_C = 4$, respectively. These two constants fill missing inference entries at the boundaries of the hierarchy. Their values are determined by considering both the treatment of missing information and the fault-detection thresholds. In the sensor diagnostic branch, the missing entry is assigned a value of 1. Saturation maps this value to 0.8, which is subsequently removed by thresholding. It therefore contributes nothing to the sensor fault indicator, leaving the diagnosis dependent on the actual residuals.

In the component diagnostic branch, missing terminal information would otherwise leave the downstream propagation pattern incomplete. A fixed completion value is therefore introduced for boundary-pattern assessment. The selected value of 4 produces a baseline response of 4 after saturation, thresholding, and convolution-kernel weighting, remaining below the component diagnostic threshold of 5. This completion alone cannot trigger an alarm: the net weighted contribution from the actual residuals must exceed 1 for the component response to exceed the threshold. Boundary completion thus supports fault identification while retaining a margin between the baseline response and the alarm threshold.

These values are selected using predefined physical residual patterns and checked against the diagnostic thresholds, providing a feasible set of boundary-completion constants.

The validation results are summarized in Table S2. The selected kernels correctly diagnose all generated residual-pattern samples, including normal samples, sensor-fault samples, and component/cascade-fault samples.

Table S2: Validation results of the optimized diagnostic kernels under standard and disturbed residual-pattern conditions.

Condition	Number of samples	Correct diagnosis	Accuracy
Normal condition	200	200	100.0%
Sensor fault	400	400	100.0%
Component/cascade fault	500	500	100.0%
Standard validation total	1100	1100	100.0%
Noisy normal condition	2000	2000	100.0%
Sensor fault with random disturbance	2000	1831	91.55%
Component fault with random disturbance	2000	1972	98.60%

Therefore, $\boldsymbol{K}_S$ and $\boldsymbol{K}_C$ are not manually tuned CNN kernels. They are obtained through a physics-constrained optimization process based on EHSS residual topology. The positive weights enhance the target residual topology, while the negative weights suppress competing fault patterns. This design is suitable for zero-sample fault diagnosis because it does not require labeled target fault samples to train a fully learned diagnostic head. The resulting kernels maintain reliable diagnostic performance under random disturbances, demonstrating their robustness to noise in fault diagnosis as shown in the Table S2.

B. Convolution-Based Multi-Level Information Fusion

Following the hierarchical signal construction and convolution-kernel design, the designed kernels are applied to extract and fuse diagnostic information across different physical levels. Specifically, the deviation matrices of the two pressure branches are processed using the sensor- and component-fault kernels to obtain local convolutional responses. These responses are then fused according to the physical hierarchy to construct diagnostic indicators for fault identification and localization.

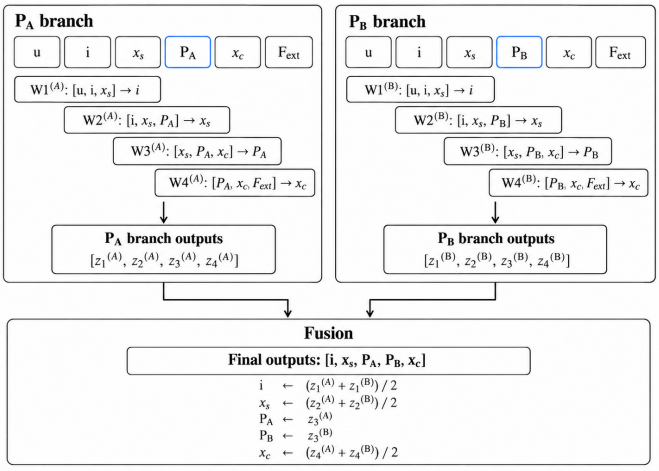


Fig. S2: Pressure-Branch Convolution and Indicator Fusion.

The chamber pressures are jointly propagated during hierarchical inference, with $\boldsymbol{P} = [P_A, P_B]^{\mathrm{T}}$. The normalized residuals of P_A and P_B are retained separately for convolutional feature extraction.

Let $\boldsymbol{d}_v \in \mathbb{R}^{1\times 3}$ denote the diagnostic matrix row associated with signal v. The internal rows contain the normalized upstream, PI-RNN, and downstream residuals, while the boundary rows follow Eq. (22) of the main text. The two pressure components are arranged into separate convolution branches, yielding

$$\boldsymbol{D}^{(A)} = \begin{bmatrix} \boldsymbol{d}_u \\ \boldsymbol{d}_i \\ \boldsymbol{d}_{x_s} \\ \boldsymbol{d}_{P_A} \\ \boldsymbol{d}_{x_c} \\ \boldsymbol{d}_{F_{\mathrm{ext}}} \end{bmatrix}, \qquad \boldsymbol{D}^{(B)} = \begin{bmatrix} \boldsymbol{d}_u \\ \boldsymbol{d}_i \\ \boldsymbol{d}_{x_s} \\ \boldsymbol{d}_{P_B} \\ \boldsymbol{d}_{x_c} \\ \boldsymbol{d}_{F_{\mathrm{ext}}} \end{bmatrix} \in \mathbb{R}^{6\times 3}. \tag{S41}$$

The two matrices share all rows except the pressure row. The shared residuals retain the coupled pressure information from the preceding hierarchical inference.

Following the preprocessing in Eq. (24) of the main text, each branch undergoes a 3×3 valid convolution with unit stride and no padding:

$$\begin{aligned} \boldsymbol{z}^{(A)} &= \mathrm{Conv}_{\mathrm{valid}}(\widetilde{\boldsymbol{D}}^{(A)}, \boldsymbol{K}) = \begin{bmatrix} z_1^{(A)} \\ z_2^{(A)} \\ z_3^{(A)} \\ z_4^{(A)} \end{bmatrix}, \\ \boldsymbol{z}^{(B)} &= \mathrm{Conv}_{\mathrm{valid}}(\widetilde{\boldsymbol{D}}^{(B)}, \boldsymbol{K}) = \begin{bmatrix} z_1^{(B)} \\ z_2^{(B)} \\ z_3^{(B)} \\ z_4^{(B)} \end{bmatrix}, \end{aligned} \qquad \boldsymbol{z}^{(A)}, \boldsymbol{z}^{(B)} \in \mathbb{R}^{4\times 1}, \tag{S42}$$

where $\widetilde{\boldsymbol{D}}^{(A)}$ and $\widetilde{\boldsymbol{D}}^{(B)}$ denote the preprocessed matrices, and $\boldsymbol{K} \in \mathbb{R}^{3\times 3}$ is the corresponding diagnostic kernel.

The four convolution windows are $[u, i, x_s]$, $[i, x_s, P_A]$, $[x_s, P_A, x_c]$, and $[P_A, x_c, F_{\mathrm{ext}}]$ for branch A, and $[u, i, x_s]$, $[i, x_s, P_B]$, $[x_s, P_B, x_c]$, and $[P_B, x_c, F_{\mathrm{ext}}]$ for branch B. They produce four responses corresponding to i, x_s, P_A (or P_B), and x_c, respectively.

For each diagnostic type $v \in \{S, C\}$, the two branch outputs are fused into

$$\boldsymbol{\mathcal{I}}_v = \begin{bmatrix} (z_1^{(A)} + z_1^{(B)})/2 \\ (z_2^{(A)} + z_2^{(B)})/2 \\ z_3^{(A)} \\ z_3^{(B)} \\ (z_4^{(A)} + z_4^{(B)})/2 \end{bmatrix} \in \mathbb{R}^{5\times 1}. \tag{S43}$$

The five entries correspond to $[i, x_s, P_A, P_B, x_c]^{\mathrm{T}}$. The shared-channel responses are averaged, whereas the two pressure responses are retained separately to preserve chamber-specific diagnostic information.

C. Fault localization

Based on the multi-level diagnostic indicators obtained through convolutional information fusion, fault localization is performed by analyzing their activation patterns along the physical hierarchy. Sensor faults are identified by localized inconsistencies, whereas component faults are characterized by deviations propagating to downstream levels. By combining these distinct patterns with the hierarchical dependencies of the EHSS, the proposed method determines the fault category and identifies its corresponding physical location. The monitored hierarchy is coil current i, spool displacement x_s, the paired chamber pressures (P_A, P_B), and piston displacement x_c. The two chamber pressures are at the same level and are interpreted jointly.

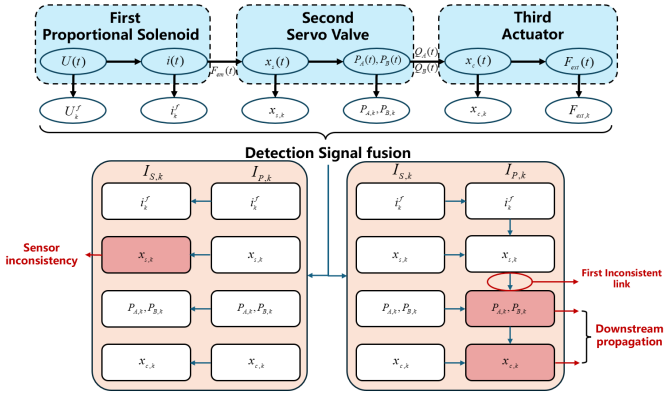


Fig. S3: Diagnostic logic for single-fault scenarios. The sensor indicator $\mathcal{I}_{S,k}$ identifies a local anomaly in an individual measurement channel, while the component indicator $\mathcal{I}_{C,k}$ identifies an inconsistency along the physical links in the causal chain.

Diagnosis proceeds along the physical causal chain of the EHSS: coil current i drives spool displacement x_s; spool motion regulates

the flow and determines the chamber pressures P_A and P_B; and these pressures drive the piston displacement x_c. The two chamber pressures are parallel signals at the same hydraulic level. The sensor indicator $I_{S,k}$ identifies a local anomaly in an individual measurement channel. For example, if $I_{S,k}$ persistently indicates an anomaly at x_s, while $I_{C,k}$ shows no corresponding downstream propagation pattern, the inconsistency is localized to the spool-displacement measurement channel and attributed to an LVDT sensor fault. By contrast, the component indicator $I_{C,k}$ is used to identify an inconsistency along the physical links in the causal chain. The signals are examined from upstream to downstream; when a level and the subsequent signals first deviate persistently from the expected pattern while the upstream signals remain consistent, the link between them is treated as the candidate fault location. In the spool-wear example, the component indicators for i and x_s remain normal, whereas the first inconsistency appears between x_s and the pressure level and then propagates to x_c. The diagnosis therefore focuses on the spool-displacement-to-flow/pressure conversion in the servo valve, consistent with spool wear, rather than on the LVDT or the hydraulic cylinder. A diagnosis is confirmed only when the relevant indicator exceeds its threshold for the required persistence duration, thereby excluding transient fluctuations.

Five-Seed Confusion Matrices and Healthy-Data False Alarms: The five seeds are 42, 123, 2024, 3407, and 5678. Each seed-specific matrix pools counts over 1, 5, 10, and 20 Hz. Per seed, there are 48,000 LVDT-fault segments after combining the two LVDT manifestations, 24,000 proportional-solenoid-fault segments, and 4,000 spool-fault segments, giving N_F = 76,000. The healthy evaluation contains 200 segments at each frequency, giving N_H = 800. These healthy segments were evaluated separately using each of the five trained models. Repeated predictions on the fixed test set do not constitute additional independent acquisitions.

No false alarms were observed among the 800 healthy segments for any of the five models (Table S3). As shown in Table S4, the proposed method achieves a mean detection rate of $95.81 \pm 0.37\%$ and a fault-only correct-diagnosis rate of $92.48 \pm 0.02\%$, with $4.19 \pm 0.37\%$ missed detections and $3.33 \pm 0.36\%$ detected but unlocalized anomalies. The correct-localization rates are $95.09 \pm 0.01\%$ for LVDT faults, $87.58 \pm 0.06\%$ for proportional solenoid degradation, and $90.60 \pm 0.13\%$ for spool wear. No inter-class misclassifications were observed among the supported fault origins. These results are based on fault-only accuracy, with standard deviations reflecting training-seed variability on fixed test data.

The absence of observed healthy false alarms and inter-class misclassifications is consistent with the physics-guided diagnostic criteria, which require fault-specific indicator patterns to persist beyond the prescribed thresholds. However, physical operating limits may suppress certain residual responses and prevent complete fault localization. For example, when the actuator reaches its stroke limit, the piston-displacement residual may remain negligible despite a component fault. Consequently, spool wear may activate the pressure indicators without activating the piston-displacement indicator, resulting in an Other/Unlocalized diagnosis rather than an incorrect fault assignment.

Unlike conventional score-based classifiers, the proposed method determines fault origins using predefined physical fault signatures, incorporating hierarchical propagation patterns, fixed thresholds, and temporal persistence. Varying these thresholds may simultaneously affect fault detection, localization, and rejection decisions, rather than simply adjusting the sensitivity of a single classification score. Therefore, a conventional single-threshold ROC curve does not directly characterize the proposed structured diagnostic rule. Instead, performance is evaluated under the prescribed decision criteria, with an observed healthy FAR of 0% and a mean detection rate of 95.81%.

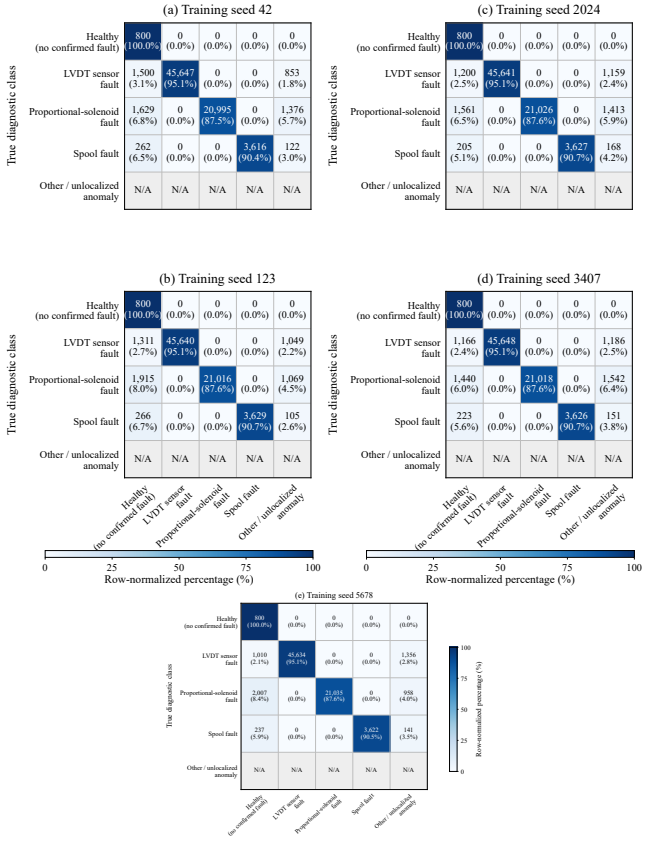


Fig. S4: PI-RNN confusion matrices for training seeds 42, 123, 2024, 3407, and 5678, pooling counts over the four excitation frequencies. The rows show the complete true diagnostic classes and the columns show the complete predicted diagnostic outcomes.

Table S3: Healthy-data evaluation. All five seed-specific models give the counts shown at each frequency.

Excitation	Healthy segments	False alarms	$\mathrm{FAR}_{\boldsymbol{H}}$ (%)	Specificity (%)
1 Hz	200	0	0.00	100.00
5 Hz	200	0	0.00	100.00
10 Hz	200	0	0.00	100.00
20 Hz	200	0	0.00	100.00
All frequencies	800	0	0.00	100.00

Table S4: PI-RNN diagnostic performance calculated from the five seed-specific confusion matrices. All rates are percentages; SD is the sample standard deviation across the five seeds.

Seed	Healthy FAR	Detection rate	Correct diagnosis	Missed detection	Unlocalized alarm
42	0.00	95.54	92.44	4.46	3.09
123	0.00	95.41	92.48	4.59	2.93
2024	0.00	96.10	92.49	3.90	3.61
3407	0.00	96.28	92.49	3.72	3.79
5678	0.00	95.72	92.49	4.28	3.23
Mean ± SD	0.00 ± 0.00	95.81 ± 0.37	92.48 ± 0.02	4.19 ± 0.37	3.33 ± 0.36

S3. Comparative Analysis of PI-RNN and EKF

A. Implementation of the EKF.

To further evaluate the effectiveness of the proposed PI-RNN model, a comparison with the extended Kalman filter (EKF) is conducted using PI-RNN one-step predictions and EKF posterior state estimates. The EKF estimates system states through model-based prediction and recursive measurement correction, whereas the proposed PI-RNN incorporates RK4-based state propagation and trainable state-dependent dynamics to capture the normal behavior of the EHSS. This comparison examines their prediction and posterior-estimation accuracy and computational efficiency, highlighting the advantages of the proposed PI-RNN for residual-based fault diagnosis. The process and measurement-noise covariance matrices are denoted by $\mathbf{\Sigma}_w$ and $\mathbf{\Sigma}_v$, respectively, to distinguish them from hydraulic flow Q_A, Q_B. The state/output orders are

$$\mathbf{x}_{k+1} = f_d(\mathbf{x}_k, u_k) + \mathbf{w}_k, \qquad \mathbf{y}_k = \mathbf{H}\mathbf{x}_k + \mathbf{v}_k, \quad \mathbf{w}_k \sim \mathcal{N}(\mathbf{0}, \mathbf{\Sigma}_w), \quad \mathbf{v}_k \sim \mathcal{N}(\mathbf{0}, \mathbf{\Sigma}_v). \tag{S44}$$

To maintain consistency with the physical variables used in the PI-RNN, the EKF state vector is defined as

$$\mathbf{x}_k = [\, i_k,\ x_{s,k},\ \dot{x}_{s,k},\ P_{A,k},\ P_{B,k},\ Q_{A,k},\ Q_{B,k},\ x_{c,k},\ \dot{x}_{c,k}\,]^{\mathrm{T}},$$

where i is the proportional-solenoid current, x_s and $\dot{x}_s$ are the spool displacement and velocity, P_A and P_B are the chamber pressures, Q_A and Q_B are the corresponding flow rates, and x_c and $\dot{x}_c$ are the actuator displacement and velocity.

The measured-output vector is

$$\mathbf{y}_k = [\, i_k,\ x_{s,k},\ P_{A,k},\ P_{B,k},\ x_{c,k}\,]^{\mathrm{T}}.$$

Therefore, because these quantities are directly selected from the state vector, the measurement model is

$$\mathbf{y}_k = \mathbf{H}\mathbf{x}_k + \mathbf{v}_k,$$

with

$$\mathbf{H} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \end{bmatrix}. \tag{S45}$$

At each sampling instant, the EKF first performs a one-step prediction,

$$\hat{\mathbf{x}}_{k+1|k} = f_d(\hat{\mathbf{x}}_{k|k}, u_k),$$

and propagates the state covariance according to

$$\mathbf{\Sigma}_{k+1|k} = \mathbf{J}_k \mathbf{\Sigma}_{k|k} \mathbf{J}_k^{\mathrm{T}} + \mathbf{\Sigma}_w,$$

where the state-transition Jacobian is defined as

$$\mathbf{J}_k = \left. \frac{\partial f_d(\mathbf{x}, u_k)}{\partial \mathbf{x}} \right|_{\mathbf{x}=\hat{\mathbf{x}}_{k|k}}.$$

Because the EHSS model contains strongly nonlinear electromagnetic, mechanical, hydraulic-flow, and pressure dynamics, the Jacobian is evaluated numerically rather than analytically. Specifically, the implementation uses a central finite-difference approximation,

$$[\mathbf{J}_k]_{:,j} = \frac{f_d(\hat{\mathbf{x}}_{k|k} + \delta_j \mathbf{e}_j, u_k) - f_d(\hat{\mathbf{x}}_{k|k} - \delta_j \mathbf{e}_j, u_k)}{2\delta_j},$$

where $\mathbf{e}_j$ denotes the j-th unit vector and δ_j is the finite-difference perturbation for the corresponding state. Neither an analytical Jacobian nor automatic differentiation is used in the EKF implementation.

Before the measurement at $k+1$ is incorporated, the corresponding one-step predicted measurement is

$$\hat{\mathbf{y}}_{k+1|k} = \mathbf{H}\hat{\mathbf{x}}_{k+1|k}.$$

Once the new measurement $\mathbf{y}_{k+1}$ becomes available, the innovation is

$$\mathbf{r}_{k+1} = \mathbf{y}_{k+1} - \hat{\mathbf{y}}_{k+1|k},$$

with innovation covariance

$$\mathbf{\Sigma}_{k+1}^{\mathrm{inn}} = \mathbf{H}\mathbf{\Sigma}_{k+1|k}\mathbf{H}^{\mathrm{T}} + \mathbf{\Sigma}_v.$$

The Kalman gain is then calculated as

$$\mathbf{G}_{k+1} = \mathbf{\Sigma}_{k+1|k}\mathbf{H}^{\mathrm{T}} \left(\mathbf{\Sigma}_{k+1}^{\mathrm{inn}}\right)^{-1},$$

and the posterior state estimate is updated according to

$$\hat{\mathbf{x}}_{k+1|k+1} = \hat{\mathbf{x}}_{k+1|k} + \mathbf{G}_{k+1}\left(\mathbf{y}_{k+1} - \mathbf{H}\hat{\mathbf{x}}_{k+1|k}\right).$$

The posterior covariance is

$$\mathbf{\Sigma}_{k+1|k+1} = (\mathbf{I} - \mathbf{G}_{k+1}\mathbf{H})\,\mathbf{\Sigma}_{k+1|k}.$$

Reorganizing the state-update equation gives

$$\hat{\mathbf{x}}_{k+1|k+1} = \underbrace{(\mathbf{I} - \mathbf{G}_{k+1}\mathbf{H})}_{\mathbf{W}_{k+1}^{\mathrm{pred}}}\hat{\mathbf{x}}_{k+1|k} + \underbrace{\mathbf{G}_{k+1}}_{\mathbf{W}_{k+1}^{\mathrm{meas}}}\mathbf{y}_{k+1}.$$

Thus,

$$\mathbf{W}_{k+1}^{\mathrm{pred}} = \mathbf{I} - \mathbf{G}_{k+1}\mathbf{H}$$

represents the contribution of the model-based prior estimate, whereas

$$\mathbf{W}_{k+1}^{\mathrm{meas}} = \mathbf{G}_{k+1}$$

determines how the new sensor measurements modify that estimate. For a directly measured scalar state, the update reduces to

$$\hat{x}_{k+1}^{+} = (1 - G_{k+1})\hat{x}_{k+1}^{-} + G_{k+1}y_{k+1}.$$

Accordingly,

$$w_{k+1}^{\mathrm{pred}} = 1 - G_{k+1}, \qquad w_{k+1}^{\mathrm{meas}} = G_{k+1}.$$

The diagonal covariance settings in the original SI coordinates (not the display units of Table S5) are

$$\begin{aligned} \mathrm{diag}(\mathbf{\Sigma}_0) &= (2.5{\times}10^{-3}, 4{\times}10^{-10}, 10^{-4}, 4{\times}10^{10}, 4{\times}10^{10}, \\ &\qquad 10^{-8}, 10^{-8}, 4{\times}10^{-8}, 10^{-4}), \\ \mathrm{diag}(\mathbf{\Sigma}_w) &= (4{\times}10^{-4}, 4{\times}10^{-12}, 2.5{\times}10^{-5}, 2.5{\times}10^{9}, 2.5{\times}10^{9}, \\ &\qquad 10^{-10}, 10^{-10}, 10^{-10}, 4{\times}10^{-6}), \\ \mathrm{diag}(\mathbf{\Sigma}_v) &= (9.38{\times}10^{-3}, 1.05{\times}10^{-9}, 1.72{\times}10^{10}, 2.10{\times}10^{11}, 8.21{\times}10^{-8}). \end{aligned} \tag{S46}$$

In the measured-output order $[\, i, x_s, P_A, P_B, x_c\,]$, the corresponding measurement-noise standard deviations in display units are approximately 0.097 A, 0.032 mm, 1.31 bar, 4.58 bar, and 0.287 mm.

B. Integration of Dynamic L_f into the EKF.

The dynamic-L_f EKF retains the state vector, observation matrix, and prediction–correction recursion of Section S3-A. The change is confined to the flow-state transition. For each chamber $j \in \{A, B\}$, the fixed inertance in the flow equation is replaced by a positive state-dependent value:

$$\dot{Q}_j = \frac{f_j(x_s, P_A, P_B) - R_{f,j}Q_j}{L_{f,j}^{\mathrm{EKF}}(\mathbf{x}, t)}, \qquad L_{f,j}^{\mathrm{EKF}}(\mathbf{x}, t) = L_{f,\min} + \mathrm{Softplus}\left[\mathcal{N}_{\boldsymbol{\theta}_L}(\boldsymbol{P}, x_s, x_c\right] \tag{S47}$$

Here f_j is the corresponding valve-flow term already defined by the physical model; no additional correction is applied to the pressure, spool, or actuator equations. The mapping parameters θ are identified using healthy data and frozen during EKF evaluation. The filter evaluates $L_{f,A}^{\mathrm{EKF}}$ and $L_{f,B}^{\mathrm{EKF}}$ at the current propagated state rather than augmenting $\mathbf{x}$ with online parameter estimates. For the 20-Hz implementation, $g_\theta(\mathbf{x}, t) = \mathbf{W}_L\mathbf{z} + \mathbf{b}_L$, where $\mathbf{z} = [\, P_A, P_B, x_s, x_c, \dot{x}_s, \dot{x}_c\,]^{\mathrm{T}}$; The archived 1-Hz EKF run used a previously fitted 14-coefficient state/phase basis. It is retained as a separate numerical comparison and is not presented as an evaluation of the final six-input affine mapping.

The state-dependent transition f_d^{dyn} is obtained by numerically integrating Eq. (S47) with the other state equations. Its Jacobian is computed by the same central-difference procedure as in Section S3-A, so each perturbed transition reevaluates the dynamic L_f. The predicted covariance and the innovation, gain, state correction, and posterior covariance then follow the EKF equations in that section. The fixed-L_f control replaces $L_{f,j}^{\mathrm{EKF}}$ by a constant in the same flow equation.

C. PI-RNN and EKF Results at 1 and 20 Hz.

The PI-RNN output is its one-step state prediction. The EKF outputs are posterior estimates after the current measurement is assimilated at both frequencies.

For each measured channel j, the reported relative error is

$$E_{j,L_2}(\%) = 100\,\frac{\|\widehat{\mathbf{y}}_j - \mathbf{y}_j^f\|_2}{\|\mathbf{y}_j^f\|_2}, \tag{S48}$$

where $\mathbf{y}_j^f$ is the filtered measurement over the indicated evaluation points and $\widehat{\mathbf{y}}_j$ is the displayed method output.

(1)1-Hz comparison: Table S5 and Figs. S5–S6 reproduce the three-method comparison: PI-RNN prediction, the originally reported fixed-L_f EKF posterior estimate, and the dynamic-L_f EKF posterior estimate. The dynamic-L_f configuration has relative L_2 errors of 5.761%, 1.649%, 2.034%, 0.934%, and 4.884% for i, x_s, x_c, P_A, and P_B, respectively.

Table S5: Relative L_2 errors (%) for the 1-Hz record. The PI-RNN column is a one-step prediction; both EKF columns are posterior estimates.

State	PI-RNN	Fixed-L_f EKF	Dynamic-L_f EKF
i	1.898	5.992	5.761
x_s	1.501	1.655	1.649
x_c	1.086	6.895	2.034
P_A	1.511	0.936	0.934
P_B	3.098	4.893	4.884
Mean	1.819	4.074	3.052

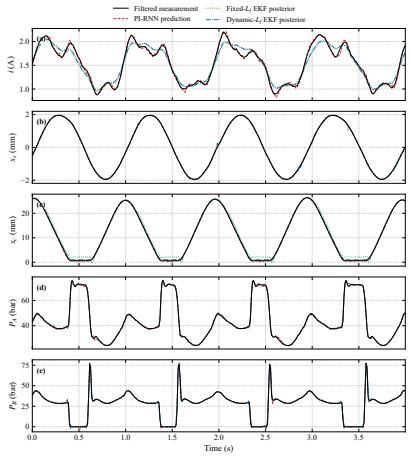


Fig. S5: Filtered measurements, PI-RNN one-step prediction, fixed-L_f EKF and dynamic-L_f EKF posterior estimates for the 1-Hz record.

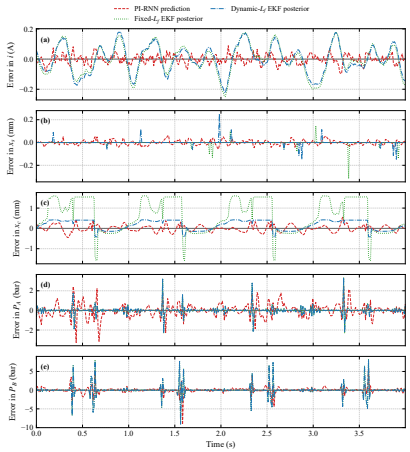


Fig. S6: PI-RNN one-step prediction errors and EKF posterior-estimation errors relative to filtered measurements for the three outputs in Fig. S5.

(2) 20-Hz comparison: At 20 Hz, Fig. S7 compares PI-RNN one-step predictions and EKF posterior estimates with the ground-truth measurements, while Fig. S8 presents the corresponding prediction and posterior-estimation errors. Table S6 summarizes the average prediction and posterior-estimation errors over the evaluation interval.

Table S6: Relative L_2 errors (%) for the 20-Hz record. The PI-RNN column is a one-step prediction; both EKF columns are posterior estimates.

State	PI-RNN	Fixed-L_f EKF	Dynamic-L_f EKF
i	0.800	2.098	1.291
x_s	3.051	8.274	7.385
x_c	7.020	10.350	10.271
P_A	2.761	3.506	3.231
P_B	14.254	20.851	20.164
Mean	5.577	9.016	8.469

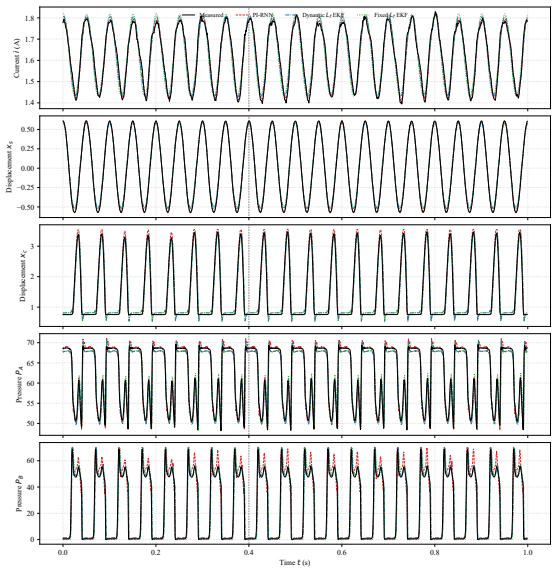


Fig. S7: PI-RNN one-step predictions and fixed-L_f and dynamic-L_f EKF posterior estimates on the 20-Hz record.

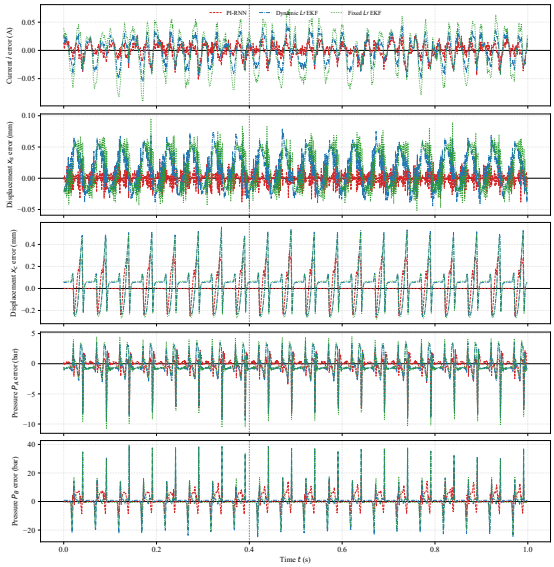


Fig. S8: PI-RNN one-step prediction errors and EKF posterior-estimation errors relative to filtered measurements for the three outputs in Fig. S7.

At 1 Hz, the hybrid PI-RNN achieves a lower mean relative L_2 error (1.819%) than the fixed-L_f EKF (4.074%) and dynamic-L_f EKF (3.052%), although both EKFs perform slightly better for P_A.

At 20 Hz, PI-RNN is evaluated using one-step predictions, while both EKFs are evaluated using posterior estimates. The hybrid PI-RNN achieves lower errors across all five monitored signals, with a mean relative L_2 error of 5.577%, compared with 9.016% and 8.469% for the fixed-L_f and dynamic-L_f EKFs, respectively.

Overall, these results demonstrate that the proposed RK4-integrated hybrid PI-RNN achieves lower mean errors in the comparisons with the evaluated EKF posterior estimates.

S4. Learnability and Identifiability of L_f

A. 1. Convergence and practical identifiability of L_f

In the proposed PI-RNN model, the hydraulic inertance parameters $L_{f,A}$ and $L_{f,B}$ are not directly measured in the experiment. However, they explicitly appear in the valve flow dynamic residuals and are constrained by the coupled pressure dynamics, reconstructed chamber flow rates, spool displacement, and actuator motion. Therefore, these parameters can be estimated indirectly by minimizing the discrepancy between the measured system response and the physics-constrained PI-RNN prediction.

To examine the effect of these trainable parameters, we first tested the case where L_f was fixed. As shown in Fig. S9, the normalized loss remained relatively large, indicating that a fixed inertance parameter was insufficient to represent the unsteady flow dynamics under the tested operating conditions.

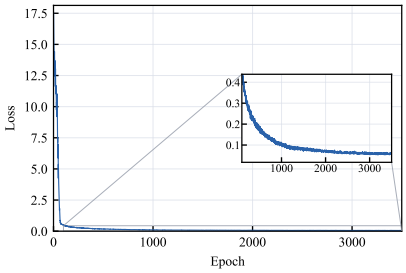


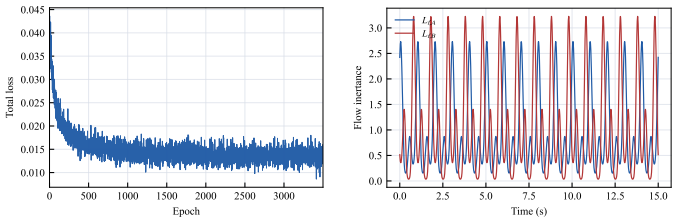
Fig. S9: Overall loss curve with fixed flow inertance parameter.

We then treated the equivalent flow inertance as a trainable state-dependent mapping in the PI-RNN framework. The mapping is the same six-input, two-output affine map followed by componentwise Softplus as in the manuscript:

$$\begin{aligned} \mathbf{z} &= [P_B, P_A, x_s, x_c, \dot{x}_s, \dot{x}_c]^{\mathrm{T}}, \\ \mathbf{L}_f(\mathbf{z}) &= L_{f,\min} + \boldsymbol{T}_j \,\mathrm{Softplus}(\mathbf{W}_L \mathbf{z} + \mathbf{b}_L), \end{aligned} \tag{S49}$$

where $\mathbf{W}_L \in \mathbb{R}^{2\times 6}$ and $\mathbf{b}_L \in \mathbb{R}^2$ contain 14 trainable scalar coefficients in total, and $L_{f,\min} > 0$ imposes a positive lower bound. The mapping parameters are fixed after training, while the evaluated $\mathbf{L}_f$ varies with the instantaneous state.

After retraining L_f as a variable parameter, the overall loss was reduced by approximately 85%, as shown in Fig. S10a. This result indicates that the hydraulic inertance term contributes meaningful information to the PI-RNN loss formulation and can be effectively estimated under the tested excitation conditions.



(a) Overall loss curve after retraining the equivalent flow inertance.

(b) Identified equivalent flow inertance $L_{f,A}(t)$ and $L_{f,B}(t)$.

Fig. S10: Retraining loss and identified equivalent flow-inertance trajectories.

The identified equivalent inertance trajectories are shown in Fig. S10b, and the corresponding trainable coefficient trajectories are shown in Fig. S11. These results show that the learned inertance terms remain bounded and improve the dynamic consistency of the PI-RNN prediction.

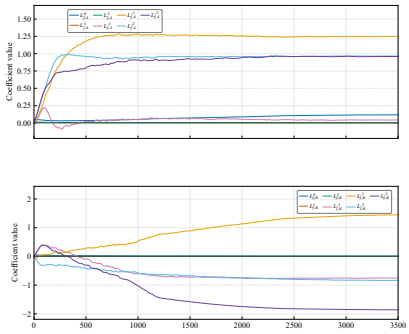


Fig. S11: Trainable coefficient trajectories for $L_{f,A}(t)$ and $L_{f,B}(t)$.

B. Convergence Analysis of L_f Under Different Initializations

We initialize L_f from several initial values and repeat the complete training procedure. Despite the different initial conditions, the estimated L_f converges to nearly the same value/function. This indicates that the final estimate is not strongly dependent on the initial guess.

Fig. S12 shows the convergence trajectories of L_f under different initial values, demonstrating that all runs gradually approach a similar solution.

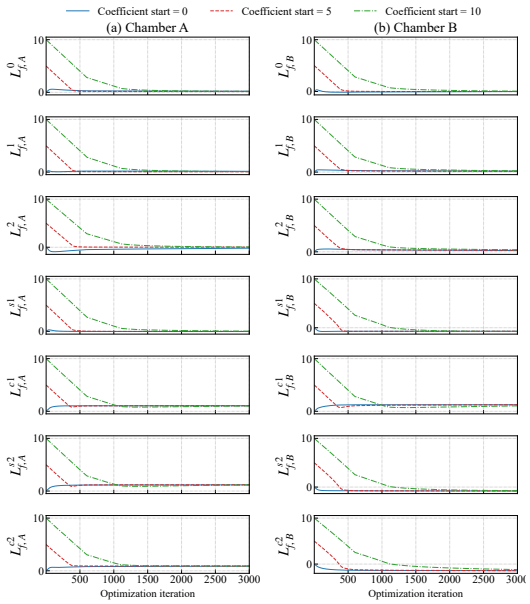


Fig. S12: Convergence trajectories of L_f under different initial values.

C. Reasons for absorbing unmodeled physical quantities

The learned L_f contain a contribution from dynamics that are not explicitly included in the reduced-order flow model. For hydraulic path $j \in \{A, B\}$, the actual flow dynamics can be written as

$$L^0_{f,j}\dot{Q}_j + R_{f,j}Q_j = f_j(x_s, P_A, P_B, Q_j) + d_j, \tag{S50}$$

where $L^0_{f,j}$ is the nominal flow-inertance parameter and d_j denotes the neglected dynamics, such as local flow transients, pressure-dependent effects, and other higher-order hydraulic behavior.

In the proposed model, these dynamics are not represented by an additional neural residual. Instead, a state-dependent effective parameter is used:

$$L^{\text{eff}}_{f,j}(\boldsymbol{h})\dot{Q}_j + R_{f,j}Q_j = f_j(x_s, P_A, P_B, Q_j). \tag{S51}$$

Comparing the two equations gives

$$\left[L^{\text{eff}}_{f,j}(\boldsymbol{h}) - L^0_{f,j}\right]\dot{Q}_j = -d_j. \tag{S52}$$

This relation shows that model discrepancies that can be represented through the transient-flow term may appear as a change in the estimated L_f. Therefore, the learned quantity should be interpreted as a *state-dependent effective flow-inertance parameter*, rather than the nominal physical inertance alone.

The effective parameter is represented as

$$L^{\text{eff}}_{f,j}(\boldsymbol{h}) = L_{f,\min} + \boldsymbol{T}_j \,\text{Softplus}\left[\mathcal{N}_{\boldsymbol{\theta}_L,j}(x_s, \boldsymbol{P}, x_c, \dot{x}_s, \dot{x}_c)\right], \tag{S53}$$

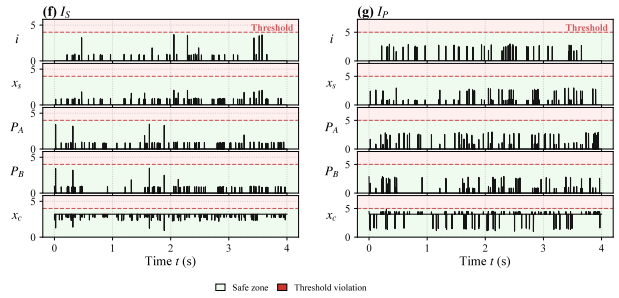
which allows L_f to vary with the hydraulic state while maintaining $L^{\text{eff}}_{f,j} > 0$.

Thus, L_f can absorb part of the unmodeled dynamics, but this compensation is introduced through the prescribed $L_f\dot{Q}_j$ term rather than through an unrestricted additive residual. The present data do not allow the nominal physical contribution and the absorbed model discrepancy to be separated uniquely.

S5. Effects of physical-parameter variation on diagnostic indicators

This section compares 20-Hz healthy-data indicators with nominal L_f and with $0.90L_f$. Fig. S13 reproduces only the diagnostic-indicator panels of that artwork. The trajectories are omitted for compactness. Model mismatch changes the responses, particularly near the actuator-displacement stage, illustrating sensitivity to the flow-model parameter rather than immunity to parameter variation.

(a) Nominal L_f



(b) $0.90L_f$

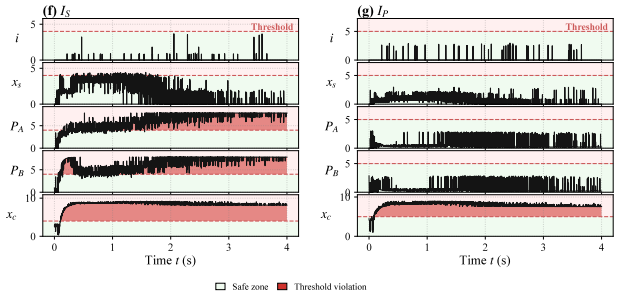


Fig. S13: Sensor indicators I_S (left) and component indicators I_C (right) under nominal and perturbed flow-inertance settings. Red dashed lines show the original thresholds. These are qualitative examples, not confirmed false-alarm-rate estimates.

An instantaneous threshold crossing is not a confirmed alarm: the $0.2T_c$ continuous-duration rule must be applied to the underlying records. The archived artwork provides no verified confirmed-alarm count or trial denominator, so a numerical healthy false-alarm rate is not inferred. Varying L_f also does not replace independent bulk-modulus and friction-variation tests. Those remain separate validation requirements.